Thesis Draft

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1 Bayesian Modelling

We observe a discrete time collection of functions $\{f_1, ..., f_t\}$, where each function describe a single day patient tremor probability.
$f_t(x)$ describes the behaviour of the t-th day as function of x.

We decompose each function as:

$$f_t(x) = \mu(x) + g_t(x) + \epsilon_t(x) \tag{1}$$

where $\mu(x)$ is the main effect, $g_t(x)$ is the autoregressive effect and $\epsilon_t(x)$ is the error term.
The **main effect** describes the structural effect, ignoring time dependence.
The **autoregressive effect** describes time dependence.
The **error term** describes the local error.

1.1 Prior and Structural Assumptions

Each function is observed a $x_1, .., x_n$, shared over time. At the generic x_i value at time t, we have:

- $\epsilon_t(x_i) \sim N(0, \sigma^2\phi_i)$, so that
$$\boldsymbol{\epsilon}_t(x) \sim N_n(\mathbf{0}, \sigma^2\Phi) \tag{2}$$
with
$$\Phi = \begin{pmatrix} \phi_1 & 0 & \cdots & 0 \\ 0 & \phi_2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & \phi_n \end{pmatrix}$$
- $\mu(x) = \sum_{j=1}^{k} \alpha_j b_j(x)$, where $\{b_1(x), ..., b_k(x)\}$ forms a basis system (e.g. splines, Fourier, ...)
- $g_t(x) = \sum_{j=1}^{k} \gamma_{tj} b_j(x)$, with $\boldsymbol{\gamma}_t = A^T \boldsymbol{\gamma}_{t-1}$

A priori, we have:

$$\sigma^2 \sim IG(a_\sigma, b_\sigma) \tag{3}$$
$$\Phi \sim IW(S_0, \nu_0) \tag{4}$$
$$\boldsymbol{\alpha} \sim N_k(\mathbf{a}_0, \Sigma_0) \tag{5}$$
$$A|\Phi \sim MN(A_0, V_l, C\Phi C^T) \tag{6}$$

1.2 Posterior Computations

First, note that the likelihood of the function is given by:

$$L(f_1, ..., f_T|\sigma^2, \Phi, \boldsymbol{\alpha}, A) = \prod_{t=1}^{T}\prod_{i=1}^{n}(2\pi\sigma^2\phi_i)^{-\frac{1}{2}} \exp\left\{-\frac{1}{2}\frac{(f_t(x_i) - \mu(x_i) - g_t(x_i))^2}{\sigma^2\phi_i}\right\} \tag{7}$$

1.2.1 Sigma

$$\pi(\sigma^2|f_1,...,f_T;\Phi,\boldsymbol{\alpha},A) \propto \pi(\sigma^2)L(f_1,...,f_T|\sigma^2,\Phi,\boldsymbol{\alpha},A) \tag{8}$$

$$\propto (\sigma^2)^{-a_\sigma-1}\exp\left\{-\frac{b_\sigma}{\sigma^2}\right\} \times (\sigma^2)^{-\frac{Tn}{2}}\exp\left\{-\frac{1}{2\sigma^2}\sum_{t=1}^{T}\sum_{i=1}^{n}\frac{(f_t(x_i)-\mu(x_i)-g_t(x_i))^2}{\phi_i}\right\} \tag{9}$$

$$= (\sigma^2)^{-(a_0-\frac{Tn}{2})-1}\exp\left\{-\frac{1}{\sigma^2}\left[b_\sigma+\frac{1}{2}\sum_{t=1}^{T}\sum_{i=1}^{n}\frac{(f_t(x_i)-\mu(x_i)-g_t(x_i))^2}{\phi_i}\right]\right\} \tag{10}$$

which identifies an Inverse-Gamma distribution with

$$\sigma^2|f_1,...,f_T;\Phi,\boldsymbol{\alpha},A \sim IG(a^*_\sigma,b^*_\sigma)$$

$$a^*_\sigma = a_\sigma + \frac{nT}{2}$$

$$b^*_\sigma = b_\sigma + \frac{1}{2}\sum_{t=1}^{T}\sum_{i=1}^{n}\frac{(f_t(x_i)-\mu(x_i)-g_t(x_i))^2}{\phi_i}$$

1.2.2 Alpha

$$\pi(\boldsymbol{\alpha}|f_1,...,f_T;\sigma^2,\Phi,A) \propto L(f_1,...,f_T|\sigma^2,\Phi,\boldsymbol{\alpha},A)\pi(\boldsymbol{\alpha})$$

$$\propto \exp\left\{-\frac{1}{2\sigma^2}\sum_{t=1}^{T}(\mathbf{f}_t - B\boldsymbol{\alpha}+\mathbf{g}_t)^T\Phi^{-1}(\mathbf{f}_t - B\boldsymbol{\alpha}-\mathbf{g}_t)\right\} \times \exp\left\{-\frac{1}{2}(\boldsymbol{\alpha}-\mathbf{a}_0)^T\Sigma_0^{-1}(\boldsymbol{\alpha}-\mathbf{a}_0)\right\}$$

where

$$B = \begin{pmatrix} b_1(x_1) & b_2(x_1) & \cdots & b_k(x_1) \\ b_1(x_2) & b_2(x_2) & \ddots & \vdots \\ \vdots & \ddots & \ddots & \\ b_1(x_n) & \cdots & & b_k(x_n) \end{pmatrix}$$

and

$$\mathbf{f}_t = \begin{pmatrix} f_t(x_1) \\ f_t(x_2) \\ \vdots \\ f_t(x_n) \end{pmatrix}$$

we also define $\mathbf{z}_t = (\mathbf{f}_t - \mathbf{g}_t)$, so we have that the previous equation becomes

$$\propto \exp\left\{-\frac{1}{2}\left[\boldsymbol{\alpha}\Sigma_0^{-1}\boldsymbol{\alpha} - 2\boldsymbol{\alpha}^T\Sigma_0\mathbf{a}_0 - 2\frac{1}{\sigma^2}\sum_{t=1}^{T}\boldsymbol{\alpha}^T B^T\Phi\mathbf{z}_t - \frac{1}{\sigma^2}\sum_{t=1}^{T}\boldsymbol{\alpha}^T B^T\Phi^{-1}B\boldsymbol{\alpha}\right]\right\}$$

$$= \exp\left\{-\frac{1}{2}\left[\boldsymbol{\alpha}^T\left(\Sigma_0^{-1}+\frac{T}{\sigma^2}B^T\Phi^{-1}B\right)\boldsymbol{\alpha} + 2\boldsymbol{\alpha}^T\left(\Sigma_0^{-1}\mathbf{a}_0 + \frac{1}{\sigma^2}\sum_{t=1}^{T}B^T\Phi^{-1}\mathbf{z}_t\right)\right]\right\}$$

which identifies a Gaussian distrbution, with

$$\boldsymbol{\alpha}|f_1,...,f_T;\sigma^2,\Phi,A \sim N_k(\mathbf{a}_T,\Sigma_T)$$

$$\Sigma_T = (\Sigma_0^{-1} + \frac{T}{\sigma^2}B^T\Phi^{-1}B)^{-1}$$

$$\mathbf{a}_T = \Sigma_T\left[\Sigma_0^{-1}\mathbf{a}_0 + \frac{1}{\sigma^2}\sum_{t=1}^{T}B^T\Phi^{-1}(\mathbf{f}_t - \mathbf{g}_t)\right]$$

1.2.3 A

Note that

$$L(f_1, ..., f_T|\sigma^2, \Phi, \boldsymbol{\alpha}, A)$$
$$\propto \exp\left\{-\frac{1}{2\sigma^2}\sum_{t=1}^{T}(\mathbf{f}_t - \boldsymbol{\mu} - BA\boldsymbol{\gamma}_{t-1})^T\Phi^{-1}(\mathbf{f}_t - \boldsymbol{\mu} - BA\boldsymbol{\gamma}_{t-1})\right\}$$
$$\text{we define } \mathbf{y}_t = (\mathbf{f}_t - \boldsymbol{\mu}_t)$$

$$= \exp\left\{-\frac{1}{2\sigma^2}\sum_{t=1}^{T}(C\mathbf{y}_t - A\boldsymbol{\gamma}_{t-1})^T(C\Phi C^T)^{-1}(C\mathbf{y}_t - A\boldsymbol{\gamma}_{t-1})\right\}$$
$$\text{with } C = (B^TB)^{-1}B^T$$
$$\propto \exp\left\{-\frac{1}{2\sigma^2}\left[(C\Phi C^T)^{-1}(\mathbf{Y} - \mathbf{\Gamma}A)^T(\mathbf{Y} - \mathbf{\Gamma}A)\right]\right\}$$

We define here

$$\mathbf{Y} = \begin{pmatrix} \mathbf{y}_1^T \\ \mathbf{y}_2^T \\ \vdots \\ \mathbf{y}_T^T \end{pmatrix}$$

and

$$\mathbf{\Gamma} = \begin{pmatrix} \boldsymbol{\gamma}_0^T \\ \boldsymbol{\gamma}_1^T \\ \vdots \\ \boldsymbol{\gamma}_{T-1}^T \end{pmatrix}$$

$$= \exp\left\{-\frac{1}{2\sigma^2}tr\left[(C\Phi C^T)^{-1}(YC^T - \Gamma\hat{A})^T(YC^T - \Gamma\hat{A})\right]\right\} \times \exp\left\{-\frac{1}{2\sigma^2}tr[(C\Phi C^T)^{-1}(A - \hat{A})^T\Gamma^T\Gamma(A - \hat{A})]\right\}$$
$$\text{then we have}$$
$$\pi(A|f_1, ..., f_T; \sigma^2, \Phi, \boldsymbol{\alpha})$$
$$\propto \exp\left\{-\frac{1}{2}tr[(C\Phi C^T)^{-1}(A - \hat{A})^T\Gamma^T\Gamma(A - \hat{A})]\right\} \times \exp\left\{-\frac{1}{2}tr[(C\Phi C^T)^{-1}(A - A_0)^TV_0(A - A_0)]\right\}$$
$$= \exp\left\{-\frac{1}{2}tr[(C\Phi C^T)^{-1}(A - A_T)^TV_T^{-1}(A - A_T)]\right\}$$

Which identifies a matrix normal

$$A|\sigma^2, \Phi, \boldsymbol{\alpha} \sim MN_{k\times k}(A_T, V_T, C\Phi C^T)$$
$$V_T = \left(V_0^{-1} + \frac{\Gamma^T\Gamma}{\sigma^2}\right)^{-1}$$
$$A_T = V_T\left(V_0^{-1}A_0 + \frac{\Gamma^TYC^T}{\sigma^2}\right)$$

1.2.4 Phi

For Φ matrix is not possible to find a conjugate posterior distribution.

1.3 Independent Metropolis-Hastings for Φ

Instead of trying to explicit Φ in a conjugate way, we try the approach Matropolis-within-Gibbs; in particular, at every iteration of the Gibbs sampler, Φ is being generated by an Independent Metropolis-Hastings iteration.
We consider the prior for Φ:

$$\Phi \sim IW(S_0, \nu_0) \tag{11}$$

At every iteration, we generate a value from the prior distribution We then compute the acceptance probability as:

$$\alpha(\Phi^*, \Phi_{t-1}) = \min\{1, \exp\{\log L(f_1, ..., f_T|\Phi^*) + \log \pi(\Phi^*) - \log L(f_1, ..., f_T|\Phi_{t-1}) - \pi(\Phi_{t-1})\}\}$$

1.3.1 Multiple Try Metropolis Hastings

The previous method is correct from a theoretical point of view, but suffers some numerical problems due to the IW density function in many dimensions being almost 0 everywhere.
To overcome this issue, the first attempt is to use a method called *Multiple Try Metropolis Hastings (MTMH)*. This method, at every iteration, follows this algorithm: Suppose the current state is Φ_{t-1}.

1. Draw $\Phi_1^*, ..., \Phi_m^*$ from the prior distribution. Compute the weights as:

$$w(\Phi_j^*, \Phi_{t-1}) = L(f_1, ..., f_T|\Phi_j^*)\pi(\Phi_j^*) \tag{12}$$

2. Select Φ^* from Φ_j^* with probability proportional to the weights.
3. Produce a reference set by drawing $\Phi_1, ..., \Phi_{m-1}$ from the prior distribution. Set $\Phi_m = \Phi_{t-1}$ (the current point).
4. Accept Φ^* with probability:

$$\alpha(\Phi^*, \Phi_{t-1}) = \min\left\{1, \frac{w(\Phi_1^*, \Phi_{t-1}) + ... + w(\Phi_m^*, \Phi_{t-1})}{w(\Phi_1, \Phi^*) + ... + w(\Phi_m, \Phi^*)}\right\} \tag{13}$$

We can see some improvements, but the density problem still holds.

1.3.2 Fourier Decomposition of Error Term

We have that

$$\Phi = Cov(\epsilon_t(x), \epsilon_t(x')) \tag{14}$$

but with a Metropolis Hastings algorithm and a very fine x grid, the posterior for Φ becomes computationally untractable. We the choose an orthonormal Fourier basis system $\{d_k(x)\}_{k=1}^K$ in order to approximate:

$$\epsilon_t(x) \approx \sum_{k=1}^{K} r_{t,k} d_k(x) \tag{15}$$

which, in a vectorial representation becomes $\epsilon_t(x) = D(x)\mathbf{r}_t$.
We also assume that $r_t \sim N(0, \Psi)$. The covariance between points x and x' becomes

$$Cov(\epsilon_t(x), \epsilon_t(x')) = Cov(D(x)\mathbf{r}_t, D(x')\mathbf{r}_t)) = D(x)\Psi D(x')^T \tag{16}$$

The MTMH algorithm in now used to generate a value from the posterior distribution of Ψ, recreating, at each iteration, the original matrix Φ.

1.4 Higher Autoregressive Order

Until now, we have just considered an autoregressive model of order one for $g_t(x)$ component. We now want to generalize to the case of multiple autoregressive order.
The model has still the form of 1, but now the definition of g changes such as:

$$g_t(x) = \sum_{j=1}^{k} \gamma_{tj} b_j(x)$$

where

$$\boldsymbol{\gamma}_t = \sum_{r=1}^{p} A_r \gamma_{t-r}$$

In doing so, we now have a number p of autoregressive matrices. The likelihood is still

$$L(f_1, ..., f_T|\sigma^2, \Phi, \boldsymbol{\alpha}, A_1, ..., A_p) = \prod_{t=1}^{T}\prod_{i=1}^{n}(2\pi\sigma^2\phi_i)^{-\frac{1}{2}} \exp\left\{-\frac{1}{2}\frac{(f_t(x_i) - \mu(x_i) - g_t(x_i))^2}{\sigma^2\phi_i}\right\} \tag{17}$$

In this new situation, we can rewrite the likelihood as:

$$L(f_1, ..., f_t|\sigma^2, \Phi, \boldsymbol{\alpha}, A_1, ..., A_p) \propto \exp\left\{-\frac{1}{2\sigma^2}\sum_{t=1}^{T}\left(\mathbf{y}_t - \sum_{j=1}^{p} BA_j\boldsymbol{\gamma}_{t-j}\right)^T \Phi^{-1}\left(\mathbf{y}_t - \sum_{j=1}^{p} BA_j\boldsymbol{\gamma}_{t-j}\right)\right\}$$

$$= \exp\left\{-\frac{1}{2\sigma^2}\sum_{t=1}^{T}\left(\mathbf{y}_t - \sum_{r=1}^{p} BA_j\boldsymbol{\gamma}_{t-j}\right)^T (C^{-1}C)^T\Phi^{-1}(C^{-1}C)\left(\mathbf{y}_t - \sum_{r=1}^{p} BA_j\boldsymbol{\gamma}_{t-j}\right)\right\}$$

$$= \exp\left\{-\frac{1}{2\sigma^2}\sum_{t=1}^{T}\left(C\mathbf{y}_t - \sum_{r=1}^{p} A_j\boldsymbol{\gamma}_{t-j}\right)^T (C\Phi C^T)^{-1}\left(C\mathbf{y}_t - \sum_{r=1}^{p} A_j\boldsymbol{\gamma}_{t-j}\right)\right\}$$

changing to the matrix notation, we get

$$= \exp\left\{-\frac{1}{2\sigma^2}\left[(C\Phi C^T)\left(YC^T - \sum_{r=1}^{p}\Gamma_r A_r\right)^T\left(YC^T - \sum_{r=1}^{p}\Gamma_r A_r\right)\right]\right\}$$

where we have defined here

$$\mathbf{Y} = \begin{pmatrix} \mathbf{y}_1^T \\ \mathbf{y}_2^T \\ \vdots \\ \mathbf{y}_T^T \end{pmatrix}$$

and

$$\Gamma_r = \begin{pmatrix} \boldsymbol{\gamma}_{\boldsymbol{p-r}}^{\boldsymbol{T}} \\ \boldsymbol{\gamma}_{\boldsymbol{p-r+1}}^{\boldsymbol{T}} \\ \vdots \\ \boldsymbol{\gamma}_{\boldsymbol{T-r}}^{\boldsymbol{T}} \end{pmatrix}$$

that means that, for every $r \in \{1, ...p\}$, we can rewrite the likelihood as:

$$\propto \exp\left\{-\frac{1}{2\sigma^2}tr\left[(C\Phi C^T)^{-1}(YC^T - \Gamma_r\hat{A}_r)^T(YC^T - \Gamma_r\hat{A}_r)\right]\right\} \times \tag{18}$$

$$\exp\left\{-\frac{1}{2\sigma^2}tr\left[(C\Phi C^T)^{-1}(A_r - \hat{A}_r)^T\Gamma_r^T\Gamma_r(A_r - \hat{A}_r)\right]\right\} \tag{19}$$

which, combined with an independent matrix normal prior for every $A_r, r \in \{1, .., p\}$, gives the conjugate posterior distribution:

$$A_r|\sigma^2, \Phi, \boldsymbol{\alpha} \sim MN_{k\times k}(A_{r,T}, V_{r,T}, C\Phi C^T)$$

$$V_{r,T} = \left(V_0^{-1} + \frac{\Gamma_r^T\Gamma_r}{\sigma^2}\right)^{-1}$$

$$A_{r,T} = V_{r,T}\left(V_0^{-1}A_0 + \frac{\Gamma_r^T YC^T}{\sigma^2}\right)$$

1.5 Spike and Slab prior for p

The aim is to identify the true autoregressive orders that significantly contribute to the functional autoregressive model. To achieve this, we introduce a spike-and-slab prior on the autoregressive coefficient matrices.

This means that the prior distribution for every A_r matrix, with $r \in \{1, ..., p\}$ is:

$$A_r|\Phi, \eta_r \sim (1 - \eta_r)MN(A_0, \tau V_0, \tau C\Phi C^T) + \eta_r MN(A_0, V_0, C\Phi C^T) \tag{20}$$

where $\tau \ll 1$. In this formulation, $\eta_r \in \{0, 1\}$ is a latent inclusion indicator, and the parameter τ controls the amount of shrinkage towards the spike component.
In this way the conjugacy still holds, so that:

$$A_r|\Phi, \eta_r, f_1, ..., f_T, \boldsymbol{\alpha}, \sigma^2, A_{-r} \sim (1-\eta_r)MN(A_0, \tau V_0, \tau C\Phi C^T) + \eta_r MN(A_0, V_0, C\Phi C^T) \tag{21}$$

If A_r is drawn from the first (spike) component, it is considered effectively zero, indicating that the corresponding autoregressive order does not contribute significantly to the model.

2 Synthetic Study

Different sets of data were used to validate the model. All the data were generated using equation 1. The number of observations per curve is $n_i = n = 200, \forall i \in \{1, ..., T\}$. A specific grid of equally spaced points for each function was defined without loss of generality, from which the respective functional data were generated.

2.1 First Simulation: no autoregressive component

In the first simulation the data were generated by a linear combination of B-splines without the autoregressive component. The Φ matrix is generated from an Inverse Wishart with moderate variance and diagonal scale matrix.
The arbitrary selected vector of true coefficients is $\boldsymbol{\alpha} = \{-2, 0.5, 1, 0, 1.5, -1, 0, 1.5, -1.5, -1, 1, 0\}$, so that:

$$f_t(x_i) = \sum_{j=1}^{k} \alpha_j B_j(x) + \epsilon_t(x_i)$$

$$= (-2, 0.5, 1, 0, 1.5, -1, 0, 1.5, -1.5, -1, 1, 0)' \begin{pmatrix} B_1(x_i) \\ B_2(x_i) \\ \vdots \\ B_k(x_i) \end{pmatrix} + \epsilon_t(x_i), \quad \text{with} \quad \epsilon_t(x_i) \sim N(0, \sigma^2 \phi_i)$$

where $(B_1(x_i), ..., B_k(x_i)'$ is the vector with the B-splines evaluated at x_i used to generate the ith point of the synthetic curves.

2.2 Second Simulation: one autoregressive order

In the second simulation, data are still generated by a linear combination of B-splines, but now we have also the autoregressive component. Φ matrix is generated from an Inverse Wishart with moderate variance and diagonal scale matrix.
The arbitrary selected vector of true coefficients is the same as before. The A matrix is randomly generated element-wise by a normal distribution, with mean 0 and standard deviation 1, adding then 0.5 on the diagonal elements for numerical stability.
Therefore data are being generated by:

$$f_t(x_i) = \sum_{j=1}^{k} \alpha_j B_j(x) + \sum_{j=1}^{k} A\Gamma_{j,t-1} B_j(x) + \epsilon_t(x_i)$$

where $(B_1(x_i), ..., B_k(x_i)'$ is the vector with B-splines evaluated at x_i used to generate the ith point of the synthetic curves. Note that the B-splines vector is the same both for $\mu(x)$ and $g_t(x)$.

2.3 Third Simulation: p autoregressive orders